# Linear actuatuators:

**Micro Linear Actuator (simple DC motor)**

I was looking for some micro linear actuators for my new project, yes, I found some, but $60 each! Excuse me! So, this is it!  About $3 each!

YXC: https://youtu.be/pYA-az9BLRI https://www.thingiverse.com/thing:2818642

BOM (100mm without limit switch)

- 1 x N20 gearbox  (3mm D shaft  length 5mm)
- 1 x M4  locknut
- 1 x M4 x 115mm screw (threaded rod)
- 1 x M4 Square nut
- 1 x 694ZZ Bearing
- threadlocker
- 8 x M3x10 screws

**Linear converter for 9gr servo**

When you need that extra low profile and the space is tight, print this linear converter for a 9gr **SH90 Micro servo**

3 parts for this mechanism, the cog is mounted on a servo head that's being cut in advance, just remove the sides and clean it up.

Omertali: https://www.thingiverse.com/thing:7067554

**Linear Servo Horn V2**

After having revisited this model recently I noticed some improvements that could make it easier to clean up and improve its reliability.

This is version 2 of a multipart accessory to any standard servo that converts it output into a linear motion. Designed, tested and 3D printed on makerbot style (FDM) printers this should work for you and your next robotics project.

ch4s3r:

**28BYJ-48 Mechanical Actuator #3**

I thought it would be cool to see what kind of Mechanical Actuators we could make using the 28BYJ-48 stepper motor and mostly 3D printed parts. The 28BYJ-48 is the motor that comes with most Arduino kits. It's a 5v dc motor with a max rpm of 15 and is capable of around 4 oz force/in of torque. You can get these motors with the driver for around $2 each.

3DPRINTINGWORLD : https://www.thingiverse.com/thing:2809285

**Nema17 Linear Actuator with Couplings**

Linear actuator for Nema17 motors with tripod mounting hole. The rack can be coupled with different holders, some examples are provided such as retroreflector, heart, and also the base holder is provided to design your own based on your needs

Use of the actuator

This is part of a mechanical actuator to generate oscillations with configurable frequency and peak amplitude. The C code for the microcontroller to mount the whole system is available at [GitHub]().

felipeparralejo: https://www.thingiverse.com/thing:6530349

**Mini actuator for NEMA17**

Hi. This work is an electric actuator for NEMA17. It's not complicated, very simple.

BOM:

- Spacer with a height of 12mm
- M3*8 bolt and M3 nut

Pull torque is about 1～2kg. If you want to know detail of torque, please watch this video!

- https://www.youtube.com/watch?v=3n_ZfC9AdOg
- https://www.youtube.com/watch?v=cNPqO2BJCOY